DS	Session	Stack	Question Information	Level 1 Challenge 49

Question description

Rathik organized technical round interview in Macrosoft for the set of computer science candidates.

The problem is to perform Implement a stack using single queue. you have to use queue data structure, the task is to implement stack using only given queue data structure.

Rathik have given the deadline of only 5 minutes to complete the problem.

Can you Help the candidates to complete the problem within the specified time limit ?

Function Description

x is the element to be pushed and s is stack

push(s, x)

1) Let size of q be s.

2) Enqueue x to q

3) One by one Dequeue s items from queue and enqueue them.

Removes an item from stack

pop(s)

1) Dequeue an item from q

Constraints

0 < n, m < N

1 < arr[i] < 1000

Input Format:

First line indicates n & m, where n is the number of elements to be pushed into stack and m is the number of pop operation need to be performed

next line indicates the n number stack elements

Output Format:

First line indicates top of the element of the stack

second line indicates the top of the element after the pop operation

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 2
1 2 3 4 5
```

EXPECTED OUTPUT

```
top of element 5
top of element 3
```

Test Case 2

INPUT (STDIN)

```
15 4
9 8 7 6 5 5 6 7 8 3 11 2 3 4 5
```

EXPECTED OUTPUT

```
top of element 5
top of element 11
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void Stack::push(int
val)
```

Test Case 2

KEYWORD

```
q.push(val)
```

Test Case 3

KEYWORD

```
void Stack::pop()
```

Test Case 4

KEYWORD

```
q.pop();
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

320

Test Case 3

NLOC

60

Code Editor c

```
#include <stdio.h>
int main()
{
```

Custom Input (stdin) T1 T2

Type Here